

WAE-DTI: Ensemble-based architecture for drug–target interaction prediction using descriptors and embeddings

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ABSTRACT

Drug Target Interaction (DTI) prediction is one of the main challenges in the pharmaceutical and drug discovery domains due to its high costs, time-consuming nature, and complexity of manual experiments required to evaluate interactions between large numbers of drugs and targets. In addition, a single drug can bind to multiple targets. In contrast, a single target can also bind to a number of drugs; this complicates the DTI task. Existing silico models often struggle with these challenges, particularly in managing diverse datasets. To address these issues, we introduce the Weighted Average Ensemble Drug–Target Interaction (WAE-DTI) model. Our approach integrates several descriptors and fingerprint representations to enhance prediction accuracy and generalization, namely atom pair fingerprint, Avalon, MACCS, MH, Morgan, RDKit, SEC, topological torsion, and LDP for drug representation, and ESM-2 for target representation. WAE-DTI employs a weighted average ensemble technique to handle diverse datasets effectively. The model demonstrates significant improvements over state-of-the-art methods, achieving an average mean squared error of 0.190 ± 0.001 on the Davis dataset, 0.127 ± 0.001 on Kiba, 0.143 ± 0.001 on DTC, 0.284 ± 0.004 on Metz, 0.308 ± 0.001 on ToxCast, and 0.934 ± 0.004 on STITCH. As for the classification task, WAE-DTI outperforms existing models with an AUPRC of 0.943 ± 0.001 on BioSNAP, 0.474 ± 0.011 on Davis, and 0.707 ± 0.005 on BindingDB. Our code is publicly available at ¹.

1. Introduction

The task of Drug–Target Interaction (DTI) is integral in the domain of drug discovery and repositioning. Drug repositioning can be defined as the use of pharmaceutical compounds for novel purposes that have failed to produce results in the past, leading to therapeutic discoveries [1]. For example, minoxidil was used mainly to treat ulcers; however, it is currently used to treat hair loss conditions [2].

The DTI task primarily attempts to assess the impact of a drug against a certain target; this impact is usually referred to as Binding Affinity (BA). BA is typically measured using the equilibrium dissociation constant (K_d) [3]. The lower the K_d value is, the stronger the bond between the ligand and its target. In machine learning, this is considered a regression problem. However, few papers tackle the DTI task as in the mentioned approach [4]. Instead, they use a threshold in which if the K_d value or any other indicator exceeds the threshold, it concludes whether there is an interaction or not; this approach is considered a binary classification task. Fig. 1 displays the interaction between a drug, which is often written in the Simplified Molecular

Input Line Entry System (SMILES) notation, and a protein, which consists primarily of intertwined amino acids. The interaction is commonly represented by the previously mentioned K_d .

Unfortunately, DTI is known to be costly and time consuming, as it requires extensive wetlab experiments to achieve conclusive results [5]. Moreover, the combinations of drugs and targets are vast, making wetlab experiments infeasible. In addition, it is difficult to analyze the intricate nature of the underlying biochemical processes. However, this problem can be mitigated by utilizing silico to infer DTI predictions. This can be done by using machine learning algorithms to address the problem; it is evident that machine learning can produce satisfactory results and directly cause a dramatic reduction in costs [6].

Deep learning may also detect off-target effects, which can have major consequences in medication development. When a drug molecule interacts with unanticipated targets, it causes undesired side effects. Deep learning models can help researchers build safer and more effective medications by predicting off-target effects.

Multiple deep learning models are formulated in an attempt to provide more accurate results based on existing research, which is

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¹ <https://github.com/tariqshaban/wae-dti>

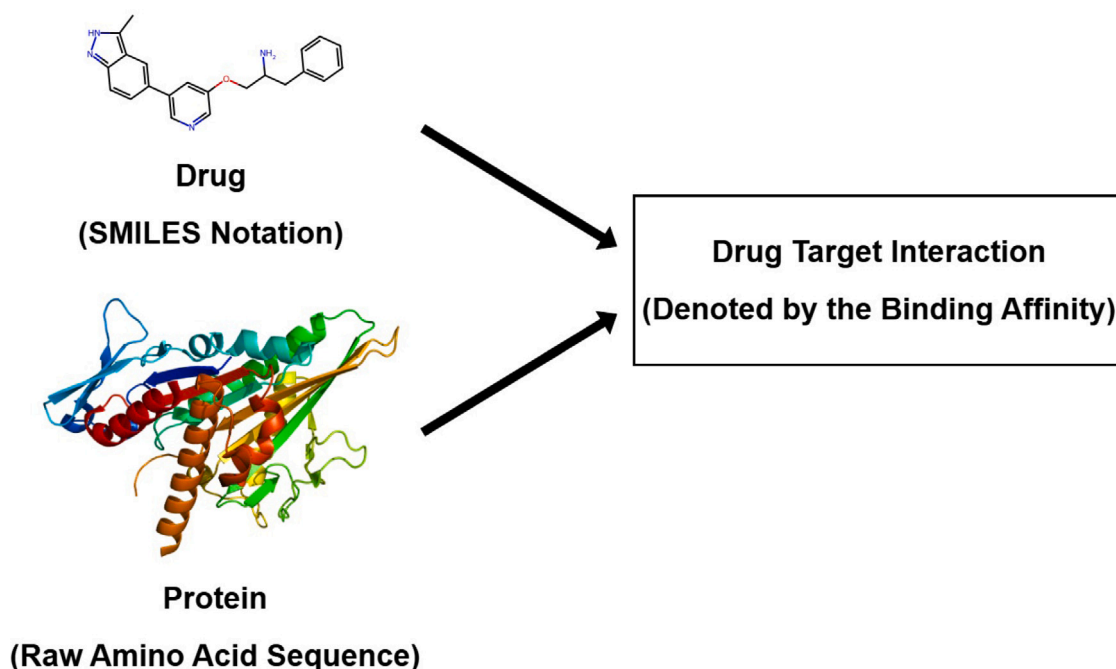


Fig. 1. DTI intuition.

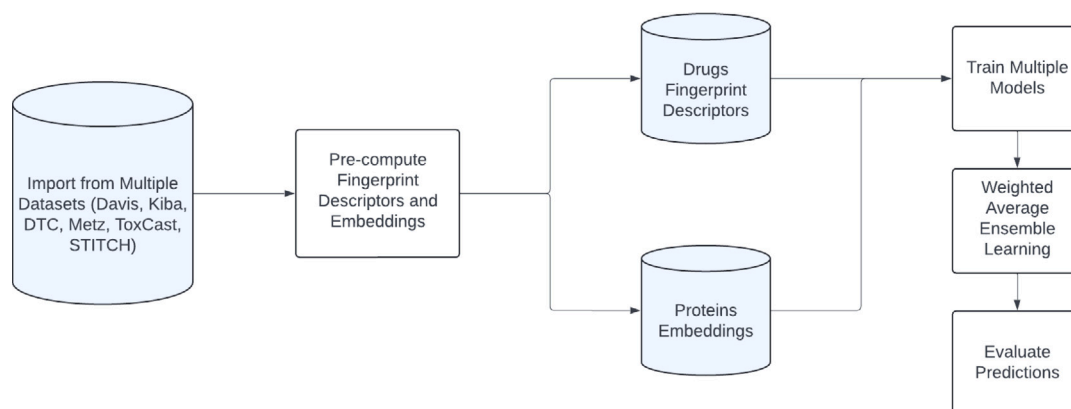


Fig. 2. The proposed model overview. After importing the datasets, the drug's and protein's representations are extracted. Then, multiple models are trained for each dataset to be weighted based on an ensemble model; the resulting model is used as a final model for evaluation.

primarily based on the mean squared error. Since the features of both proteins and targets are represented as textual data, two spaces need to be established: one for drugs and one for targets (proteins); this is commonly achieved through feature extraction, or more specifically, by using embedding. After training the models, an ensemble learning model is used to predict the binding strength; Fig. 2 outlines the basic steps taken to present the deep learning model scheme.

In the remainder of the study, a detailed discussion of related work in the DTI prediction area is provided, along with the materials and methods used to overcome the challenges of the DTI prediction task. The results are then highlighted and compared with recent studies. Finally, the findings are discussed and limitations found during the research are outlined. The aim is to effectively measure the bond strength between a drug and a protein.

1.1. Motivation

Manual experiments and laboratory tests are considered expensive to carry out because of the high costs of equipment, compound synthesis, financial resources, time, and workforce required. In addition, testing combinations of drugs and targets results in a monumental number of experiments that need to be conducted. Such issues can be resolved by using computed predictions to estimate the strength of an interaction in an efficient, timely manner.

1.2. Research questions

This study addresses the following research questions:

- **Model Performance:** Can a deep learning model be formulated to achieve better results than existing state-of-the-art models?

Specifically, how would the model perform compared to existing studies with respect to Mean Squared Error (MSE), Concordance Index (CI), and r_m^2 ?

- **Feature Representation:** Would using multiple feature representations of the drug enhance the model's performance?
- **Regression vs. Classification:** Would assuming the problem definition as classification rather than regression be sufficient? Can binding interactions be effectively treated as binary states?
- **Computational Efficiency:** How long does it take to train a model on each dataset?
- **Novel Examples:** How does the model perform when predicting unseen drugs or proteins?

1.3. Contributions

The main drawback of previous studies on drug–target interaction prediction has been the limited use of a sufficient number of datasets for training. Having an adequate number of examples boosts the model's ability to generalize well. Each dataset represents a different distribution of drugs and proteins, with some specializing in the human kinome and others focusing on drugs with small molecule representation. This diverse representation requires a model that can handle various structures of drugs and proteins. Our contribution is formulating a deep learning ensemble model that automates the prediction of the interaction between a drug and a target based on a wide range of diverse datasets. Our proposed model surpasses the performance of current state-of-the-art models on multiple dataset sources.

2. Related work

This section provides a comprehensive review of related work within the DTI interaction field, covering a spectrum from classical machine learning methods and matrix factorization to the implementation of deep learning algorithms. Most research papers treated the DTI prediction task as a classification problem, while few papers assumed that the problem is a regression task. The articles are grouped according to the methodology used to tackle the DTI prediction problem. The papers in each section are reviewed chronologically, discussing the early papers and then advancing to the recent discoveries. This section encompasses subsections, including classical machine learning models, matrix factorization and graph models, clustering and semi-supervised models, and deep learning models. Tables 1, 2, 3, and 4 summarize the related work.

2.1. Classical machine learning techniques

In the domain of classical machine learning techniques applied to DTI prediction, several studies have explored a variety of models and datasets. In particular, Support Vector Machine (SVM) has been a popular choice among researchers. For example, Faulon et al. [7] utilized an SVM classifier with a kernel function, leveraging the Kyoto Encyclopedia of Genes and Genomes (KEGG) database and benchmarking against the Predictive Toxicology Challenge (PTC). Similarly, Tabei et al. [8] combined logistic regression and SVM models, using the golden dataset along with the PubChem database for chemical fingerprints and the UniProt database for genomic information. Ding et al. [19] also employed SVM, focusing on feature selection with the golden dataset. Yu et al. [9] enhanced data quality by integrating Random Forest (RF) and SVM, representing drug–target pairs as tensor products with data from the DrugBank dataset. Furthermore, Cao et al. [10] incorporated Molecular Access System (MACCS) substructure fingerprints into an SVM model, utilizing the golden dataset.

Beyond SVM, other machine learning techniques have also been applied. For example, Laarhoven et al. [11] employed a Weighted Nearest Neighbor (WNN) algorithm in combination with a proposed

method named GIP, using the golden dataset. Wang et al. [12] utilized a restricted Boltzmann machine model with the Manually Annotated Targets And Drugs Online Resource (MATADOR) database and the Search Tool for Interactions of Chemicals (STITCH) dataset. Yuan et al. [13] developed the DrugE-Rank method, which combines feature-based and similarity-based approaches with learning to rank (LTR), using data from the DrugBank dataset and focusing on FDA-approved or experimental drugs and human protein targets. Ezzat et al. [14] addressed class imbalance using ensemble methods primarily composed of decision trees with the BindingDB dataset. Furthermore, Wang et al. [15] represented drug–target interaction pairs by incorporating the structural properties of drugs and the evolutionary information of proteins, using a random forest classifier and the golden dataset.

Additional studies have introduced various methodologies. Yao et al. [16] used molecular substructure fingerprints for molecular representation and naive Bayes for classification with the BindingDB dataset, and deployed their model on a web server. He et al. [17] proposed the SimBoost model, which is a trained gradient-boosting regression model that uses feature vectorization as input. Rayhan et al. [18] proposed the iDTI-ESBoost model, which incorporates evolutionary features and utilizes the golden dataset. Li et al. [20] developed the Discriminative Vector Machine (DVM) classifier using the golden dataset. Finally, Shi et al. [21] introduced the LRF-DTIs method, which combines the Pseudo Position-Specific Scoring Matrix (PsePSSM) with FP2 molecular fingerprinting for drug and target extraction. It also uses the Least Absolute Shrinkage and Selection Operator (LASSO) for dimensionality reduction and the Synthetic Minority Oversampling Technique (SMOTE) to address data imbalance. The method uses a random forest classifier to predict DTIs using both the DrugBank approved dataset and the gold standard dataset.

2.2. Matrix factorization and graph techniques

Numerous studies have made significant advances in the prediction of DTI using matrix factorization and graph models. Yamanishi et al. [22] introduced the concept of projecting drugs and targets into a “pharmacological space” using a bipartite graph learning method with datasets from KEGG BRITE, BRAunschweig ENzyme Database (BRENDA), SuperTarget, and DrugBank, collectively known as the “gold standard dataset”. This foundational work was further explored by Bleakley et al. [23] through bipartite graph inference with local models. Extending these methods, Yamanishi et al. [24] predicted pharmacological effects and inferred unknown DTIs using supervised bipartite graph inference with data from KEGG DRUG, KEGG LIGAND, JAPIC, KEGG GENES, and KEGG BRITE.

Several studies have focused on creating models to enhance the results from the golden dataset. Laarhoven et al. [25] paired the Gaussian interaction profile kernel with the regularized least squares classifier, and Gönen et al. [26] employed kernelized Bayesian matrix factorization. Chen et al. [27] used Network-based Random Walk with Restart on the Heterogeneous (NRWRH) network, while Mei et al. [28] integrated Neighbor-based Interaction-profile Inferring (NII) into the Bipartite Local Model (BLM-NII). Alaimo et al. [29] proposed a Network-Based Inference (NBI) model to address the limitations of pure ‘local’ models.

Cobanoglu et al. [30] utilized probabilistic matrix factorization with the DrugBank dataset, and Zheng et al. [6] introduced Multiple Similarities Collaborative Matrix Factorization (MSCMF). Yamanishi et al. [31] developed a model based on bipartite graph inference and pair-wise kernel regression, which was later deployed as a Web server. Pahikkala et al. [32] proposed the Kronecker Regularized Least Squares (KronRLS) method, a similarity-based model that is considered as a generalized model of the Regularized Least Squares (RLS) model. Liu et al. [33] introduced Neighborhood Regularized Logistic Matrix Factorization (NRLMF), and Nascimento et al. [34] extended the KronRLS model to KronRLS-MKL. Hao et al. [35] and Ezzat et al. [36] developed

Table 1
Related work comparison in classical machine learning techniques.

Paper	Dataset	Task	Methodology	Results
[7]	Predictive Toxicology Challenge dataset KEGG database DRUGBANK (test dataset)	Classification	SVM	Primarily above 80%
[8]	DrugBank	Classification	logistic regression SVM	Pairwise cross validation: SCCA: AUC:0.7975. Block-wise cross validation: SCCA: AUC: 0.7496.
[9]	DrugBank STITCH (test dataset) SuperTarget (test dataset) KEGG (test dataset)	Classification	Forest (RF) Support Vector Machine (SVM)	Concordance: 82.83% Sensitivity: 81.33% Specificity: 93.62% AUC: 90.55
[10]	KEGG database The gold standard dataset	Classification	SVM	Enzyme: Accuracy:90.31 Ion channel: Accuracy:88.91 GPCR: Accuracy:84.68 Nuclear receptor: Accuracy:83.74
[11]	The gold standard dataset	Classification	Weighted Nearest Neighbor Profile	Enzyme: AUC:0.983 Ion channel: AUC:0.986 GPCR: AUC:0.972 Nuclear receptor: AUC:0.958
[12]	MATADOR STITCH	Classification	Restricted Boltzmann machine	AUPR: 89.6
[13]	DrugBank dataset	Classification	Learning to rank Ensemble learning	Data-2: AUPR: 0:2031 Data-3: AUPR: 0:2831 Data-5: AUPR: 0:0997
[14]	DrugBank	Classification	Ensemble learning Decision trees	AUC: 0.900
[15]	The gold standard dataset	Classification	Rotation Forest	Enzyme: Accuracy:91.3 GPCR: Accuracy:84.1 Ion Channel: Accuracy:89.1 Nuclear Receptor: Accuracy:71.1
[16]	BindingDB	Classification	Naive Bayes Random forest	CPDBAS:0.746 $\pm$ 0.003 EPAFHM:0.713 $\pm$ 0.004 FDAMDD:0.807 $\pm$ 0.002
[17]	Davis Kiba Metz	Regression	Gradient boosting	Davis: RMSE:0.247 Kiba: RMSE:0.204 Metz: RMSE:0.166
[18]	The gold standard dataset	Classification	AdaBoost	Enzyme: AUPR:0.680 GPCR: AUPR:0.480 Ion Channel: AUPR:0.500 Nuclear Receptor: AUPR:0.790
[19]	The gold standard dataset	Classification	Support Vector Machine (SVM) Feature Selection (FS)	Enzyme: AUPR:0.899 GPCR: AUPR:0.821 Ion Channel: AUPR:0.929 Nuclear Receptor: AUPR:0.655
[20]	The gold standard dataset	Classification	Discriminative Vector Machine	Enzyme: Accuracy:93.16 GPCR: Accuracy:89.37 Ion Channel: Accuracy:91.73 Nuclear Receptor: Accuracy:92.22
[21]	Gold standard dataset	Classification	Lasso Random forest	Enzyme: AUC:0.9982 GPCR: AUC:0.9918 Ion Channel: AUC:0.9965 Nuclear Receptor: AUC:0.9559

models that integrate regularized least squares and matrix factorization, respectively.

Further innovations included topology-only algorithm proposed by Lu et al. [37] using the MATADOR database, Hao et al. [38] Dual Network Integrated Logistic Matrix Factorization (DNILMF), and Wu et al. [39] Substructure–Drug–Target Network-Based Inference (SDTNBI) using ChEMBL and DrugBank. Luo et al. [40] introduced DTINet for identifying new drug indications from a heterogeneous network. Olayan et al. [41] applied a non-linear similarity fusion method with a random forest classifier on a heterogeneous graph. More recently, Ding et al. [42,43] advanced the field with models such as Dual Laplacian Regularized Least Squares (DLapRLS) with

Hilbert–Schmidt Independence Criterion-based Multiple Kernel Learning (HSIC-MKL) and Fuzzy Bipartite Local Model (FBLM) using multiple kernel learning.

2.3. Clustering and semi-supervised techniques

Regarding projecting and clustering drugs and proteins into a unified plane, Shi et al. [45] devised a model that addresses missing interactions in most of the data, including non-structural and non-sequence-based information, they introduced the concept of a “super-target” based on the intuition that similar drugs interact with similar targets, which calculates the similarity measure using the golden dataset.

Table 2

Related work comparison in matrix factorization and graph techniques.

Paper	Dataset	Task	Methodology	Results
[22]	The gold standard dataset (consists of KEGG BRITE, BRENDA, SuperTarget and DrugBank)	Classification	Nearest profile Weighted profile Bipartite graph learning	(Nearest profile/Weighted profile/Bipartite graph learning) Enzyme: AUC:0.767/0.812/0.904 Ion channel: AUC:0.751/0.811/0.851 GPCR: AUC:0.729/0.739/0.899 Nuclear receptor: AUC:0.710/0.626/0.843
[23]	The gold standard dataset	Classification	Bipartite local models	Enzyme: AUC:97.6 Ion channel: AUC:97.3 GPCR: AUC:95.5 Nuclear receptor: AUC:88.1
[24]	The gold standard dataset	Classification	Supervised bipartite graph inference	Enzyme: AUC:0.892 Ion channel: AUC:0.812 GPCR: AUC:0.827 Nuclear receptor: AUC:0.835
[25]	The gold standard dataset	Classification	Regularized least squares paired with Kronecker product kernel (RLS-Kron)	AUPR: 92.7
[26]	The gold standard dataset	Classification	Kernelized Bayesian matrix factorization with twin kernels	Enzyme: AUC:0.832 Ion channel: AUC:0.799 GPCR: AUC:0.857 Nuclear receptor: AUC: 0.824
[27]	The gold standard dataset	Classification	Network-based Random Walk with Restart on the Heterogeneous network (NRWRH)	Enzyme: AUC:0.955 Ion channel: AUC:0.948 GPCR: AUC:0.976 Nuclear receptor: AUC:0.876
[28]	The gold standard dataset	Classification	Neighbor-based interaction-profile inferring (NII) with bipartite local model	Enzyme: AUC:98.8 Ion channel: AUC:99.0 GPCR: AUC:98.4 Nuclear receptor: AUC:98.1
[29]	The gold standard dataset	Classification	Network-based inference with domain-based knowledge	Enzyme: AUC: 0.9995 Ion channel: AUC: 0.9973 GPCR: AUC: 0.9995 Nuclear receptor: AUC: 1.0000
[30]	The gold standard dataset DrugBank	Classification	Probabilistic Matrix Factorization	Enzyme: AUC:0.861 Ion channel: AUC:0.904 GPCR: AUC:0.771 Nuclear receptor: AUC:0.650 DrugBank: AUC:0.794
[6]	The gold standard dataset	Classification	Multiple Similarities Collaborative Matrix Factorization (MSCMF)	Enzyme: AUPR:0.673 Ion channel: AUPR:0.773 GPCR: AUPR:0.937 Nuclear receptor: AUPR:0.894
[31]	KEGG	Classification	Bipartite graph inference Bi- partite local model Pair-wise kernel	Pair-wise cross validation: AUC:0.936 $\pm$ 0.010 Block-wise cross validation: AUC:0.870 $\pm$ 0.004
[32]	Davis Kiba	Regression	Kronecker Regularized Least Squares (KronRLS)	Davis: MSE:0.379 Kiba: MSE:0.411
[33]	The gold standard dataset	Classification	Logistic matrix factorization Neighborhood regularization	Enzyme: AUC:0.987 Ion channel: AUC:0.989 GPCR: AUC:0.969 Nuclear receptor: AUC:0.950
[34]	The gold standard dataset	Classification	Bipartite networks Multiple Kernel Learning algorithm	Enzyme: AUPR:0.8818 Ion channel: AUPR:0.8769 GPCR: AUPR:0.6440 Nuclear receptor: AUPR:0.5368
[35]	The gold standard dataset	Classification	Regularized least squares integrating Nonlinear kernel fusion (RLS-KF) Recalculated kernel matrix	Enzyme: AUPR:0.915 Ion channel: AUPR:0.925 GPCR: AUPR:0.853 Nuclear receptor: AUPR:0.909
[36]	The gold standard dataset	Classification	Weighted K nearest known neighbors Graph regularized matrix factorization	Enzyme: AUPR:0.807 Ion channel: AUPR:0.815 GPCR: AUPR:0.615 Nuclear receptor: AUPR:0.500
[37]	MATADOR database	Classification	Similarity-based algorithms (Common neighbors, Jaccard index, Preferential attachment, Katz index)	Common Neighbors: AUPR:0.3715 Jaccard Index: AUPR:0.3697 Preferential Attachment: AUPR:0.0022 Katz Index ($\mu = 0.005$): AUPR:0.3652 Katz Index ($\mu = 0.01$): AUPR:0.3486 Katz Index ($\mu = 0.02$): AUPR:0.3030

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Table 2 (continued).

Paper	Dataset	Task	Methodology	Results
[38]	The gold standard dataset	Classification	Profile kernel matrix Similarity diffusion Dual network logistic matrix factorization	Enzyme: AUPR:0.922 GPCR: AUPR:0.812 Ion Channel: AUPR:0.938 Nuclear Receptor: AUPR:0.751
[39]	ChEMBL database DrugBank database BindingDB database	Classification	Substructure-drug- target network-based inference	GPCRs: AUC:0.966 Kinases: AUC:0.958 ICs: AUC:0.971 NRs: AUC:0.932
[40]	Drug nodes: DrugBank Protein nodes: HPRD Disease nodes: Comparative Toxicogenomics Database Side-effect nodes: SIDER DTI/DDI: DrugBank	Classification	Random walk with restart Diffusion component analysis	AUROC:0.91 AUPR:0.93
[41]	Gold standard dataset DrugBank_FDA (test dataset)	Classification	Heterogeneous graph Non-linear similarity fusion	Enzyme: AUPR:52% GPCR: AUPR: 56% Ion Channel: AUPR:50% Nuclear Receptor: AUPR:71% DrugBank FDA: 35%
[42]	Gold standard dataset	Classification	Fuzzy bipartite local model (FBLM) Least squares support vector machine (LS-SVM) Multiple kernel learning (MKL)	Enzyme: AUPR: 0:898 $\pm$ 0:010 GPCR: AUPR: 0:917 $\pm$ 0:011 Ion Channel: AUPR: 0:743 $\pm$ 0:023 Nuclear Receptor: AUPR: 0:547 $\pm$ 0:101
[43]	Gold standard dataset	Classification	Dual Laplacian Regularized Least Squares Multiple Kernel Fusion	CV1: (Enzyme: AUPR:0.90, Ion Channel:AUPR:0.93, GPCR:AUPR:0.767, Nuclear Receptor:AUPR:0.561) CV2: (Enzyme: AUPR:0.423, Ion Channel:AUPR:0.417, GPCR:AUPR:0.418, Nuclear Receptor:AUPR:0.444) CV3: (Enzyme: AUPR:0.822, Ion Channel:AUPR:0.812, GPCR:AUPR:0.631, Nuclear Receptor:AUPR:0.357)

Table 3

Related work comparison in clustering and semi supervised techniques.

Paper	Dataset	Task	Methodology	Results
[44]	The gold standard dataset DBSI (test dataset)	Classification	Semi-supervised method named NetCBP	Enzyme: AUC:0.8075 Ion channel: AUC:0.8029 GPCR: AUC:0.8022 Nuclear receptor: AUC:0.7578
[45]	The gold standard dataset	Classification	Pairwise similarity Clustering	Enzyme: AUC:0.832 Ion channel: AUC:0.724 GPCR: AUC:0.933 Nuclear receptor: AUC:0.878

Chen et al. [44] attempted to rank drugs and targets by relevance by creating an architecture called NetCBP, which uses a semi-supervised learning method, they used the golden dataset as well.

2.4. Deep learning techniques

Deep learning models have also been used in the DTI task. For instance, Hu et al. [46] employed a deep learning model called Multi-scale Features Deep Representations (MFDR), which integrates autoencoders and SVM, and validated their approach using the golden dataset. Similarly, Wen et al. [47] utilized Extended Connectivity FingerPrints (ECFP) and Protein Sequence Composition (PSC) descriptors fed into a Deep Belief Network (DBN) model, drawing insights from the Drugbank database. Wang et al. [48] introduced a stacked autoencoder combined with a rotation forest, and evaluated it using the golden dataset. On the other hand, Oztürk et al. [49] assumed the input to be a raw one-dimensional array for each drug and protein, the input was passed through a label encoder and an embedding layer before being fed into a Convolutional Neural Network (CNN) model, the authors then flattened and merged the output of each CNN into a fully connected neural network, they referred to the architecture as DeepDTA. Oztürk et al. [50] later expanded their model and proposed the WideDTA architecture, which leverages word-based sequence representation rather than character-based sequence representation.

Zhao et al. [51] aimed to improve the results of DeepDTA Oztürk et al. [49] by modifying and fine-tuning the architecture, resulting in the AttentionDTA model.

Using another approach, Lim et al. [52] explored the application of graph neural networks with a distance-aware attention algorithm, employing diverse datasets such as the Directory of Useful Decoys-Enhanced (DUD-E) and PDBbind. Wan et al. [53] proposed NeoDTI, a model focused on topology-preserving representations of drugs and targets, which was trained on a dataset formulated by Luo et al. [40]. Meanwhile, Lee et al. [54] implemented DeepConv-DTI to capture local residue patterns, using datasets from DrugBank, KEGG, and the International Union of Basic and Clinical Pharmacology (IUPHAR). Torng et al. [55] implemented an unsupervised graph autoencoders and Graph-CNN for interaction prediction on multiple datasets, including druggable pockets, DUD-E, and ChEMBL. Shin et al. [56] indicated that classical CNNs cannot capture long-distance relationships; therefore, they proposed a transformer-based approach called MT-DTI, using character-embedded transformer layers for drugs and character-embedded CNN layers for proteins.

In addition, Beck et al. [57] directed their efforts towards identifying high-affinity drugs for SARS-CoV-2 proteins, comparing methods such as AutoDock Vina and MT-DTI on datasets from DTC and BindingDB, with viral sequences sourced from the National Center for Biotechnology Information (NCBI). Further diversifying the landscape,

Table 4

Related work comparison in deep learning.

Paper	Dataset	Task	Methodology	Results
[46]	The gold standard dataset	Classification	Stacked auto-encoder SVM	Enzyme: AUROC:0.969 GPCR: AUROC:0.904 Ion Channel: AUROC:0.933 Nuclear Receptor: AUROC:0.886Enzyme: AUROC:0.969
[47]	DrugBank	Classification	Deep belief network (DBN)	TPR: 0.8227 TNR: 0.8953 Accuracy: 0.8588 AUC: 0.9158
[48]	Gold standard dataset	Classification	Rotation forest Stacked Autoencoder	Enzyme: Accuracy:94.41 GPCR: Accuracy:86.69 Ion Channel: Accuracy:91.16 Nuclear Receptor: Accuracy:80.56
[49]	Davis Kiba	Regression	Embedding CNN	Davis: MSE:0.261 Kiba: MSE:0.194
[50]	Davis Kiba	Regression	Embedding CNN Word-based sequence representation	Davis: MSE:0.262 Kiba: MSE:0.179
[51]	Davis Kiba	Regression	Embedding CNN Enhanced fine-tuning	Davis: MSE:0.245 Kiba: MSE:0.162
[52]	DUD-E PDBbind	Classification	Protein Structural Information Embedding Graph neural network	DUD-E: AUROC:0.968 PDBbind:AUROC:0.935
[53]	Drugbank dataset HPRD dataset Comparative Toxicogenomics Database SIDER database	Classification	Node level embedding from heterogeneous network	AUPR:0.82 (positive/negative ratio is 0.1)
[54]	DrugBank	Classification	CNN	Accuracy: 0.79
[55]	DUD-E data set Maximum Unbiased Validation dataset	Classification	Graph-CNN	DUD-E: AUC: 0.886. MUV: AUC: 0.619.
[56]	Davis Kiba	Regression	Character-embedded transformer layers Character-embedded CNN layers	Davis: MSE:0.245 Kiba: MSE:0.193
[57]	Drug Target Common (DTC) BindingDB	Regression	Consistence-score-based averaging algorithm	N/A
[58]	DrugBank database	Classification	Semi-bipartite graph formulation Deep Neural Network	AUROC: 93%
[59]	Davis Kiba	Regression	Semi-supervised generative adversarial network (GAN) CNN	Davis: MSE:0.276 Kiba: MSE:0.224
[60]	Davis Kiba	Regression	Graph attention network (GAT) Bi-directional Gated Recurrent Unit (BiGRU) CNN	Davis: MSE:0.252 Kiba: MSE:0.193
[4]	Kiba Davis BindDB (test dataset)	Regression	BiLSTM CNN Differential evolution	(MSE/AUPR) Kiba: 0.179/0.785 Davis: 0.235/0.722 BindDB: 0.483/0.761
[61]	HIPPIE database DrugBank The gold standard dataset	Classification	graph convolutional network deep neural network	AUC: 0.93-0.98 AUPR: 0.85- 0.98
[62]	Davis Kiba	Regression	CNN Graph Convolutional Network (GCN) Graph Attention Network (GAT) Graph Isomorphism Network (GIN)	Davis: MSE:0.229 KIBA: MSE:0.139
[63]	Gold standard dataset	Classification	Cascade deep forest	Enzyme: AUC:0.975 GPCR: AUC:0.98 Ion Channel: AUC:0.995 Nuclear Receptor: AUC:0.98

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Table 4 (continued).

Paper	Dataset	Task	Methodology	Results
[64]	Davis	Regression	GCN	(MSE/CI/R _m ²)
	KIBA		LSTM	Davis: 0.232/0.895/0.680
	DTC			KIBA: 0.133/0.897/0.792
	Metz			DTC: 0.149/0.895/0.841
	ToxCast			Metz: 0.294/0.810/0.640
	STITCH			ToxCast: 0.313/0.919/0.565 STITCH: 0.992/—/0.430
[65]	Human	ClassificationRegression	Multiscale graph neural network (MGNN)	(MSE/CI/R _m ²)
	C. elegans			Davis: 0.207/0.900/0.710
	Davis		Multiscale convolutional neural network (MCNN)	KIBA: 0.128/0.902/0.801
	KIBADTC			Metz: 0.265/0.822/0.701
	Metz			ToxCast: 0.176/0.902/0.430
	ToxCast			(Precision/Recall/AUC) Human: 0.955/0.956/0.983 C. elegans: 0.980/0.967/0.991
[66]	Davis	Regression	ESM-2	(MSE/CI/R _m ²)
	KIBA		Transformer-GCN	Davis: 0.222/0.896/0.723
	DTC			KIBA: 0.135/0.895/0.780
	Metz			DTC: 0.171/0.881/0.830
	ToxCast			Metz: 0.292/0.810/0.682
	STITCH			ToxCast: 0.316/0.918/0.572 STITCH: 0.986/0.745/0.461

Manoochehri et al. [58] modeled DTI as a semi-bipartite graph and employed deep neural networks on datasets curated by Luo et al. [40]. Zhao et al. [59] utilized a semi-supervised Generative Adversarial Network (GAN) called GANsDTA to tackle the DTI prediction task, the architecture contains two GANs, one for training the drugs and the other for the targets, both of which are then combined into a CNN model. Lin et al. [60] presented DeepGS, which leverages Graph Attention Network (GAT) for drug representation, Bidirectional Gated Recurrent Unit (BiGRU) for local chemical context, and CNN for proteins, specifically, Smi2Vec and Prot2Vec were used for the vectorization process. Mahdaddi et al. [4] approached DTI identification as a regression problem, deploying CNN and attention-based Bidirectional Long Short-Term Memory (BiLSTM) on datasets including Kiba, Davis, and samples from BindingDB.

In addition, Zhao et al. [61] constructed a Drug-Protein Pair (DPP) network, integrating Graph Convolutional Network (GCN) and deep neural network models using data from the Human Integrated Protein-Protein Interaction rEference (HIPPIE), DrugBank, and the golden dataset. To address challenges related to false positives, Chu et al. [63] introduced DTI-CDF based on the Cascade Deep Forest (CDF) model, utilizing the golden dataset. Meanwhile, Nguyen et al. [62] proposed GraphDTA, which uses CNN for protein representation training and employs multiple graph-based representations for drug training, namely GCN, Graph Attention Network (GAT), Graph Isomorphism Network (GIN), and a combined GAT-GCN. To leverage feature extraction across multiple representations, Yang et al. [65] built a model comprising a Multiscale Graph Neural Network (MGNN) for drugs and a Multiscale Convolutional Neural Network (MCNN) for proteins, they also employed eXplainable Artificial Intelligence (XAI) to highlight atoms and bonds of high importance. Recently, Mukherjee et al. [64] aims to identify FDA-approved drugs that bind to SARS-CoV-2 viral proteins by developing the DeepGLSTM regression-based architecture, which primarily consists of GCN for drugs and LSTM for targets. Similarly, Daga et al. [66] proposed the LEP-AD regression architecture, which leverages transformer-based GCN and ESM-2 across diverse datasets, including Davis, Kiba, DTC, Metz, ToxCast, and STITCH. The mentioned diverse methods reveal the complexity of DTI prediction and illustrate how integrating various machine learning and deep learning techniques across different datasets can enhance the understanding of drug-target interactions.

Although existing studies have achieved promising results in DTI prediction, they primarily focus on classification tasks, predicting whether an interaction occurs or not. This study addresses DTI prediction as a regression problem, aiming to estimate interaction strength

while retaining crucial information. Furthermore, previous studies often utilize a limited number of datasets, with some restricted to a single data source. In contrast, this work leverages a broader range of datasets, encompassing six sources for the regression task and three additional sources for the classification task, to enhance the generalizability and robustness of the model. This paper intends to outperform current techniques in predicting DTI for both regression and classification tasks on several datasets.

3. Methodology

This section outlines the used datasets and discusses the methodology for performing the regression task for the binding affinity prediction. In addition, the section specifies the hardware specifications used to train and evaluate the model.

3.1. Data source

We have used six datasets to train and evaluate the models, namely Davis [67], KIBA [68], Drug Target Commons (DTC) [69], Metz [70], Toxicity Forecasting (ToxCast) [71], and Search Tool for Interactions of Chemicals (STITCH) [72].

The Davis [67] dataset includes the interactions between kinase inhibitors and kinases, covering more than 80% of the human catalytic protein kinome, with the dissociation constant (K_d) as the dependent variable. While Kiba [68] contains complementary information from various bioactivity types, including K_d , the inhibitory constant (K_i), and the half maximal inhibitory concentration (IC_{50}), which is defined as the 'KIBA score'. DTC [69] is a web platform that offers layers of abstraction for knowledge extraction, it primarily includes drug-target bioactivity data, pK_i was extracted from this dataset; which is the negative logarithm (base 10) of the K_d , using the pK_i instead of the K_d provides a more intuitive interpretation. Metz [70] also focused on kinase activity data with pK_i representing the binding affinity. ToxCast [71] is a vitro High-Throughput Screening (HTS) data for thousands of chemicals, containing hundreds of thousands of interactions. The half maximal effective concentration (AC_{50}) is the dependent variable for the dataset. STITCH [72] is a multi-source dataset that comprises a large number of interactions and uses its own scoring system.

Table 5 represents the unique number of drugs and proteins of each dataset. Also, it includes the number of experiments (interactions) between them, along with the data splitting ratios. We have collected

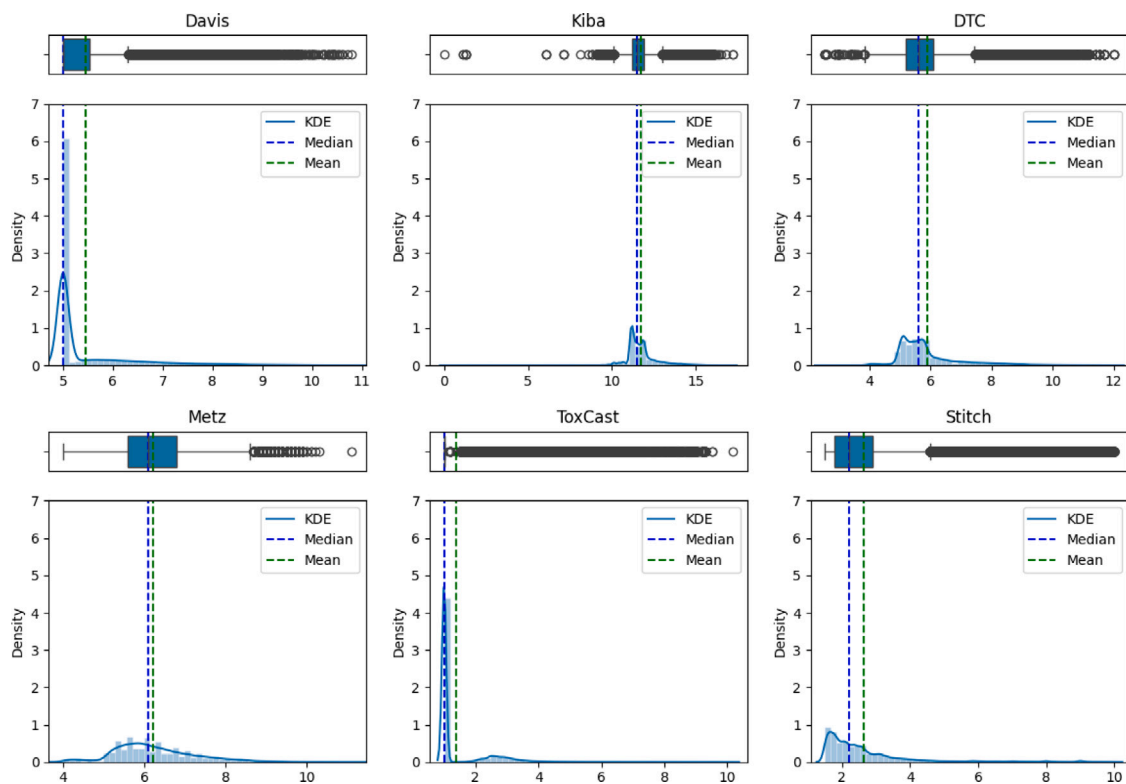


Fig. 3. Dataset distribution.

Table 5

Dataset comparison.

Dataset	Drugs	Targets	Interactions	Validation split ^a	Test split
Davis (K_d)	68	379	30 056	20%	16.66%
Kiba	2068	229	118 254	20%	16.66%
DTC (pK_i)	5983	118	67 894	20%	20%
Metz (pK_i)	1471	170	35 307	20%	30%
ToxCast (AC_{50})	7656	328	342 691	20%	20%
Stitch	165 366	14 631	1 259 388	20%	35%

^a Validation set split is deducted from the training set, not the whole dataset.

the datasets from [66] to ensure fairness in comparison, which already splits the dataset into training and testing sets; we have added a validation set to balance between fitting the training data and generalizing to unseen data. Fig. 3 represents the distribution of the dependent variable in each dataset using Kernel Density Estimation (KDE). As indicated, the Davis dataset uses K_d as an indicator for binding, as opposed to DTC and Metz, which use pK_i , ToxCast uses the AC_{50} , while Kiba and STITCH defined their customized scoring.

K_d is a form of equilibrium constant that quantifies the dissociation of a larger object into smaller components, which is the inverse of the association constant. While K_i measures the binding affinity between an inhibitor and an enzyme. K_d and K_i can be approximately equal in certain scenarios; the main difference is that K_d is a more generalized term compared to K_i . Some literatures apply the negative logarithmic transformation on both constants to form pK_d and pK_i . IC_{50} is the concentration required to achieve 50% inhibition; calculating the K_i requires finding IC_{50} , whereas AC_{50} determines the concentration of a drug needed to produce 50% of its maximum effect or activity.

Eq. (1) specifies the pK_d , where $[L]$ denotes the concentration of the free drug (unbound ligand), $[P]$ denotes the concentration of the free target (unbound protein), and $[LP]$ is the concentration of the bound complex.

$$pK_d = -\log_{10} K_d = -\log_{10} \left(\frac{[L][P]}{[LP]} \right) \quad (1)$$

While Eq. (2) calculates the pK_i ; having $[S]$ as the concentration of substrate, K_m is the Michaelis–Menten; which is the substrate concentration at which half maximum velocity is observed, the calculation of K_i originates from [73].

$$pK_i = -\log_{10} K_i = -\log_{10} \left(\frac{IC_{50}}{1 + \frac{[S]}{K_m}} \right) \quad (2)$$

3.2. Methods

The DTI prediction task consists of first vectorizing drugs and proteins into fingerprints and embeddings. Then, after acquiring the extracted features' representations, the results will be fed into a deep neural network that predicts the binding affinity value. Fig. 4 displays the model architecture in detail.

Before training, we extract the drug's features using multiple vectorization techniques and transformers to capture different aspects of the drug and improve generalizability. Feature extraction techniques include atom pair fingerprint, Avalon fingerprint, Molecular ACCess System (MACCS) keys fingerprint, MinHash (MH) fingerprint, Morgan fingerprint, RDKit fingerprint, SMILES Extended Connectivity (SEC) fingerprint, topological torsion fingerprint, and Local Degree Profile (LDP) fingerprint. All descriptors we used are from the scikit-mol [74] Python package, while the LDP model implementation is from the Karate Club [75] library.

As for the protein embedding, we have used the Evolutionary Scale Modeling (ESM-2) [76,77] for feature extraction, specifically, the model with 3 billion parameters, 36 layers, and 2560 embedding dimensions.

We have trained the model for up to 3000 epochs with an early stopping patience of 200 epochs; that is, the training process would stop if the loss did not improve within 200 consecutive epochs. Adequate computational resources and a sufficient number of examples allowed for a batch size of 1024, with a learning rate of 0.0005. To mitigate overfitting, we have used the Adam optimizer with a weight decay

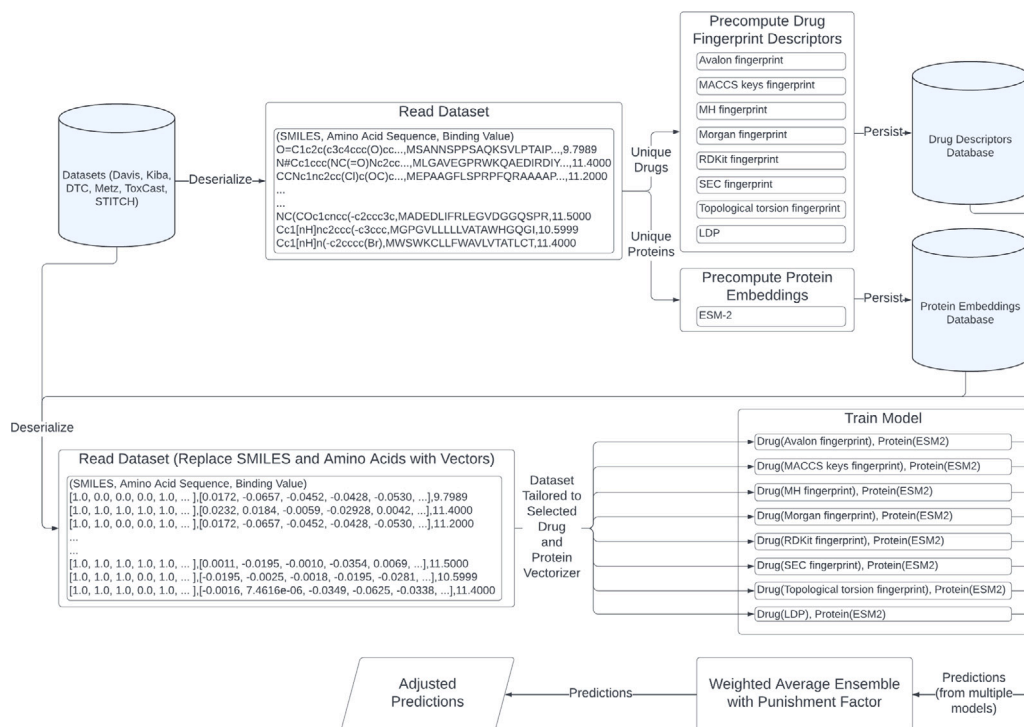


Fig. 4. The WAE-DTI architecture. First, we iterate through each dataset and extract unique drugs and proteins. For each drug, we calculate its feature representation using multiple descriptors, including atom pair fingerprint, Avalon fingerprint, MACCS keys fingerprint, MH fingerprint, Morgan fingerprint, RDKit fingerprint, SEC fingerprint, topological torsion fingerprint, and LDP. For each protein, we calculate its embedding using ESM-2. All of the features are persisted by storing them on the disk. During the training process, we use the calculated features to replace the original SMILES notations and amino acid sequences. Then, we permute each descriptor and train its isolated model. Based on the model's performance on the validation set, we leverage a weighted average ensemble with a punishment factor so that well-performing models are rewarded; conversely, poorly-performing models are punished. The output of the ensemble model is regarded as the final prediction.

of 0.00001. We have used the Mean Squared Error (MSE) as the loss function. As a measure of regularization, we inserted dropout layers with a probability of 20% between each hidden layer to randomly deactivate neurons.

We have trained each drug transformer separately. We have then used the resulting trained models as input for a weighted average ensemble model, which assigns weights to the models based on their validation set MSE. The formula for evaluating the weight of a model is expressed in Eq. (3), where w_i is the weight of model i , mse is the loss function on the validation set, p is a scalar denoting the punishment factor, and M denotes the number of input models for the weighted average ensemble. The punishment factor is set to 6 to significantly reduce the weight of a model if it is not performing well.

$$w_i = \frac{1}{mse^p} \quad \text{for } i = 1, 2, \dots, M \quad (3)$$

3.3. Computational resources

The training process was carried out on an HPE ProLiant DL380 Gen10 server, equipped with dual Intel(R) Xeon(R) Gold 6130 CPUs @ 2.10 GHz processors, 256 GB of RAM, and dual Tesla P40 graphics cards with 24 GB of virtual memory each. The model was built using PyTorch version 2.2.1 and Python version 3.11.8. The environment was set up with CUDA toolkit version 12.4 and cuDNN library version 8.9 to facilitate GPU training. The operating system used was Ubuntu version 24.04.6.

3.4. Model evaluation

To assess the model's performance, we have compared the predictions to the ground truth of the test set using Mean Squared Error (MSE), Concordance Index (CI), and r_m^2 as the evaluation metrics.

MSE is expressed as the average squared difference between the predicted values and the actual values for each example. Eq. (4) represents the mathematical notation for this metric; lower MSE indicates a less average distance between predicted and actual values.

$$MSE = \frac{1}{n} \cdot \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

CI is an extended version of the area under the ROC curve (AUC) that considers censored data. It provides a comprehensive evaluation of the model's discriminatory power. CI can be calculated using Equation (5).

$$CI = \frac{\sum_{i < j} W_{ij} I(Y_i < Y_j, T_i < T_j)}{\sum_{i < j} W_{ij}} \quad (5)$$

r_m^2 [78] is a variant of the well-known coefficient of determination (r_m) and is measured by Eq. (6) and used as a validation of regression-based quantitative structure–activity relationship (QSAR) models, specifically in the pharmaceutical domain.

$$r_m^2 = r^2 \cdot (1 - \sqrt{r^2 - r_0^2}) \quad (6)$$

4. Results

For each dataset, we extract the representations of drugs and proteins, with the drugs represented by nine different fingerprint descriptors and the proteins represented using ESM-2. These representations are extracted and persisted before training, so instead of recalculating the embeddings at every epoch, they are directly deserialized from the local disk. Next, for each dataset and drug descriptor, we start training the model by entering the selected drug descriptor paired with the ESM-2 protein embedding, the results for each model without applying

Table 6

Model performance on five experiments compared to related work for Davis and Kiba datasets.

Model	Davis (K_d)			Kiba		
	MSE	CI	r_m^2	MSE	CI	r_m^2
KronRLS ^a	0.379	0.871	0.407	0.411	0.782	0.342
SimBoost ^a	0.282	0.872	0.644	0.222	0.836	0.629
DeepDTA	0.261	0.878	0.630	0.194	0.863	0.673
MT-DTI (w/o FT)	0.268	0.875	0.633	0.220	0.844	0.584
MT-DTI	0.245	0.887	0.665	0.152	0.882	0.738
WideDTA	0.262	0.886	0.633 ^b	0.179	0.875	0.675 ^b
GANsDTA	0.276	0.881	0.653	0.224	0.866	0.675
Attention-DTA	0.216	0.893	0.677	0.155	0.882	0.755
DeepGS	0.252	0.882	0.686	0.193	0.860	0.684
GraphDTA (GCN)	0.254	0.880	–	0.139	0.889	–
GraphDTA (GAT-GCN)	0.245	0.881	–	0.139	0.891	–
GraphDTA (GAT)	0.232	0.892	0.662 ^b	0.179	0.866	0.671 ^b
GraphDTA (GIN)	0.229	0.893	0.649 ^b	0.147	0.882	0.684 ^b
CNN-AbiLSTM	0.235	0.897	0.668	0.179	0.876	0.733
DeepGLSTM	0.232	0.895	0.680	0.133	0.897	0.792
LEP-AD (AlphaFold)	0.228	0.894	0.692	0.133	0.893	0.752
LEP-AD	0.222	0.896	0.723	0.135	0.895	0.780
WAE-DTI	0.190	0.915	0.745	0.127	0.899	0.778

^a Referred from [60].^b Referred from [66].**Table 7**

Model performance on five experiments compared to related work for DTC and Metz datasets.

Model	DTC (pK_i)			Metz (pK_i)		
	MSE	CI	r_m^2	MSE	CI	r_m^2
GraphDTA (GCN) ^a	0.317	0.878	0.812	0.317	0.801	0.620
GraphDTA (GAT-GCN) ^a	0.200	0.857	0.790	0.333	0.795	0.602
GraphDTA (GAT) ^a	0.195	0.859	0.788	0.393	0.775	0.549
GraphDTA (GIN) ^a	0.176	0.876	0.798	0.317	0.800	0.645
DeepGLSTM	0.149	0.895	0.841	0.294	0.810	0.640
LEP-AD	0.171	0.881	0.830	0.292	0.810	0.682
WAE-DTI	0.143	0.898	0.839	0.284	0.813	0.676

^a Referred from [66].

ensemble learning can be seen in Appendix A. After finishing the training process, we have implemented a weighted average ensemble model on the same dataset for all models to improve the prediction generalization.

Tables 6, 7, and 8 display the mean performance obtained from WAE-DTI on the datasets, based on five repeated experiments, and compared with related studies. It can be noted that the model significantly outperformed similar studies across all datasets. Specifically, for the Davis dataset, we achieved an MSE of 0.190 ± 0.001 , a CI of 0.915 ± 0.001 , and an r_m^2 of 0.745 ± 0.003 , compared to the best-performing model (LEP-AD), which recorded an MSE of 0.222, a CI of 0.896, and an r_m^2 of 0.723. For the Kiba dataset, our model achieved an MSE of 0.127 ± 0.001 , a CI of 0.899 ± 0.000 , and an r_m^2 of 0.778 ± 0.003 , as opposed to LEP-AD, which produced an MSE of 0.135, a CI of 0.895, and an r_m^2 of 0.780. Additionally, for the DTC dataset, our model recorded an MSE of 0.143 ± 0.001 , a CI of 0.898 ± 0.001 , and an r_m^2 of 0.839 ± 0.008 , compared to LEP-AD results of 0.149 in MSE, 0.895 in CI, and 0.841 in r_m^2 . Furthermore, our model outperformed the baseline model in the Metz dataset by achieving an MSE of 0.284 ± 0.004 , a CI of 0.813 ± 0.001 , and an r_m^2 of 0.676 ± 0.006 , compared to the best-performing baseline model, which achieved an MSE of 0.289, a CI of 0.810, and an r_m^2 of 0.682. Finally, our model also delivered better results than the baseline model for both the ToxCast and STITCH datasets. Specifically, for ToxCast, we obtained an MSE of 0.308 ± 0.001 , a CI of 0.922 ± 0.000 , and an r_m^2 of 0.581 ± 0.003 , while for STITCH, we achieved an MSE of 0.934 ± 0.004 , a CI of 0.772 ± 0.001 , and an r_m^2 of 0.488 ± 0.003 . In contrast, the target paper reported results of 0.316 in MSE, 0.918 in CI, and 0.572 in r_m^2 for ToxCast, and 0.986 in MSE, 0.745 in CI, and 0.461 in r_m^2 for STITCH.

On all of the tested datasets, our method consistently achieved lower MSE and higher CI and r_m^2 compared to LEP-AD. This suggests that our approach offers improved accuracy, agreement, and prediction power on these datasets. However, for the r_m^2 metric on the Kiba, DTC, and Metz datasets, LEP-AD and DeepGLSTM reported slightly higher results. To further visualize the results, we have constructed a residual plot for each dataset, shown in Fig. 5; having the trend line aligned with the horizontal line provides a positive indication that the model fits well. In most datasets, the trend line has a slope close to zero. However, in the STITCH dataset, the trend line does not align well with the horizontal line, implying that the prediction task on STITCH is more challenging compared to other datasets, as indicated by its MSE, CI, and r_m^2 . Also, the colors of the points denote the KDE using Gaussian kernels.

In general, these results highlight the effectiveness of our proposed method, WAE-DTI, demonstrating noticeable performance across all evaluated datasets compared to the target paper.

4.1. Model performance on classification task

Although assuming the binding between a drug and a target as a binary state can be less robust than quantifying the binding strength, comparing the proposed model's results in such a task proves the agility in handling both scenarios, in addition to verifying the performance of the model in the classification task. Naturally, this requires using a different type of datasets. Moreover, the model architecture might need slight modifications, particularly to the loss function used.

4.1.1. Materials

To ensure that the work can be evaluated and compared fairly with related papers, we have used the material from the target article [79],

Table 8

Model performance on five experiments compared to related work for ToxCast and STITCH datasets.

Model	ToxCast (AC_{50})			STITCH		
	MSE	CI	r_m^2	MSE	CI	r_m^2
GraphDTA (GCN) ^a	0.316	0.917	0.561	1.022	–	0.424
GraphDTA (GAT-GCN) ^a	0.317	0.917	0.566	–	–	–
GraphDTA (GAT) ^a	0.346	0.911	0.535	–	–	–
GraphDTA (GIN) ^a	0.324	0.915	0.555	–	–	–
DeepGLSTM	0.313	0.919	0.565	0.992	–	0.430
LEP-AD	0.316	0.918	0.572	0.986	0.745	0.461
WAE-DTI	0.308	0.922	0.581	0.934	0.772	0.488

^a Referred from [66].**Table 9**

Classification dataset comparison.

Dataset	Drugs	Proteins	Training interactions		Validation interactions		Testing interactions	
			Positive	Negative	Positive	Negative	Positive	Negative
BioSNAP	4510	2181	9619	9619	1374	1374	2748	2748
Davis	68	379	1 043	1 043	160	2 846	303	5 708
BindingDB	10 665	1413	6334	6334	927	5717	1905	11 348

Table 10

Classification model hyperparameters.

Hyperparameter	Value
Epochs	3000
Early Stopping Patient	200
Early Stopping Score Determinant	AUPRC
Learning Rate	5e–4
Optimizer	Adam
Weight Decay	1e–6
Dropout Probability	20%
Loss Function	Binary cross entropy

which encompasses three datasets, MINER DTI from BioSNAP [80], Davis [67], and BindingDB [81]. Although the latest BioSNAP update contains 5017 drugs, 2324 targets, and 15 138 total interactions, an older snapshot was used to match the datasets with those in [79]. This older version includes 4510 drugs, 2181 targets, and 13 741 total interactions. Since BioSNAP comprises only positive examples, negative unseen examples were added to create a balanced dataset. Davis consists of 68 drugs and 379 targets, with 11 103 interactions that cover more than 80% of the human kinome. As for BindingDB, it consists of 10 665 drugs and 1413 targets, with 32 601 interactions. Both Davis and BindingDB measure the binding affinity using the dissociation constant K_d . To convert predictions from continuous values to binary values, any K_d less than 30 indicates a successful binding.

Table 9 specifies the number of unique drugs, unique proteins, and interactions for each dataset. The splitting of the datasets into training, validation, and testing is outlined by [79], so the comparisons of the results are valid and sound. The table also displays the distribution of binary labels for each data split.

4.1.2. Methods

The model attempts to predict whether or not a combination of a drug and a target will bind. The architecture is highly similar to the proposed regression model. However, there are subtle differences between them. Both models utilize similar embeddings and descriptors to represent drugs and proteins. Table 10 lists the hyperparameters used in the training process. The main difference is the early stopping determinant, where the regression model uses MSE, while the classification model uses the Area Under the Precision-Recall Curve (AUPRC). Also, the regression model uses MSE as its loss function, while the classification model uses the Binary Cross Entropy (BCE) loss.

Another notable difference is the ensemble weight assigned to the models; specifically, the classification model uses AUPRC. The computation of these weights is calculated using Eq. (7), where w_i denotes the weight of the model i , $auprc$ represents the model loss on the validation

set, p is a scalar that penalizes lower performance, and M is the total number of models included in the ensemble. The punishment factor p is set at 6 to significantly diminish the influence of underperforming models.

$$w_i = \text{auprc}^p \quad \text{for } i = 1, 2, \dots, M \quad (7)$$

4.1.3. Model evaluation

To compare our results with existing papers, we compare the predictions to the ground truth of the test set using multiple evaluation metrics, such as sensitivity, specificity, Area Under the ROC Curve (AUC), and Area Under the Precision-Recall Curve (AUPRC).

Sensitivity measures the probability of a positive test result. High sensitivity indicates that the test effectively identifies when the condition is present, minimizing false negatives. It is also referred to as the true positive rate. The formula for sensitivity can be expressed by (8).

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (8)$$

Specificity measures the probability of an absent (negative) test result. High specificity indicates that the test effectively identifies when the condition is truly absent, minimizing false positives. It is also referred to as the true negative rate. Specificity can be evaluated using Eq. (9).

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (9)$$

AUC measures the model's ability to distinguish between positive and negative classes by evaluating the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR). It is approximated using the trapezoidal method, as seen in Eq. (10). However, directly calculating AUC may result in an overly-optimistic estimation [82,83].

$$\text{AUC} \approx \sum_{i=1}^{n-1} \frac{(\text{Sensitivity}_i + \text{Sensitivity}_{i+1}) \cdot (\text{Specificity}_i - \text{Specificity}_{i+1})}{2} \quad (10)$$

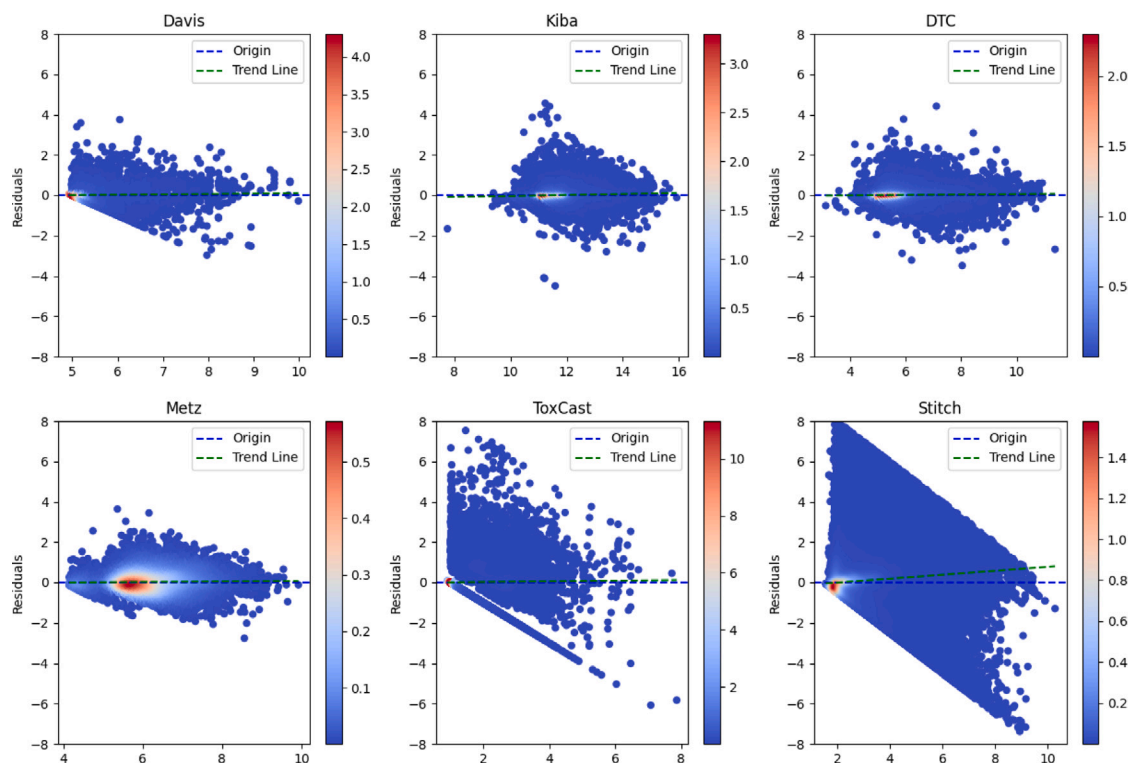


Fig. 5. Residual plots of WAE-DTI predictions on each dataset.

Table 11

Model performance on five experiments compared to related work for BioSNAP classification dataset.

Paper	Sensitivity	Specificity	AUC	AUPRC	Threshold
LR	0.755 ± 0.039	0.800 ± 0.018	0.846 ± 0.004	0.850 ± 0.011	0.434
DNN	0.776 ± 0.040	0.838 ± 0.024	0.849 ± 0.003	0.855 ± 0.010	0.499
GNN-CPI	0.780 ± 0.014	0.819 ± 0.012	0.879 ± 0.007	0.890 ± 0.004	0.349
DeepDTI	0.789 ± 0.027	0.845 ± 0.017	0.876 ± 0.005	0.876 ± 0.006	0.347
DeepDTA	0.781 ± 0.015	0.824 ± 0.012	0.876 ± 0.005	0.883 ± 0.006	0.466
DeepConv-DTI	0.770 ± 0.023	0.832 ± 0.016	0.883 ± 0.002	0.889 ± 0.005	0.441
MolTrans	0.775 ± 0.032	0.851 ± 0.014	0.895 ± 0.002	0.901 ± 0.004	0.431
WAE-DTI	0.862 ± 0.003	0.871 ± 0.006	0.935 ± 0.001	0.943 ± 0.001	0.478

AUPRC is similar to AUC. However, AUPRC evaluates the trade-off between precision and recall rather than TPR and FPR. It can also be roughly calculated using the trapezoidal method, as indicated in Eq. (11).

$$AUPRC \approx \sum_{i=1}^{n-1} \frac{(Precision_i + Precision_{i+1}) \cdot (Recall_{i+1} - Recall_i)}{2} \quad (11)$$

4.1.4. Results

After carrying out the training process and evaluating the models based on the testing set, Appendix B denotes the results for each model before applying ensemble learning. Also, Fig. 6 illustrates the model performance in terms of AUC and AUPRC. In addition, we have compared the predictions with related papers based on sensitivity, specificity, AUC, and AUPRC. The proposed model proved to obtain significant improvement in the DTI classification task. As shown in Tables 11, 12, and 13, which represent the mean and standard deviation of the evaluation metrics of the model after applying the ensemble voting on five random runs, so as to ensure that the performance is not the result of pure chance. Such results are also compared with relevant and state-of-the-art models, namely LR [79], DNN [79], GNN-CPI [84], DeepDTI [47], DeepDTA [49], DeepConv-DTI [54], and MolTrans [79].

The selected threshold for determining the trade-off between sensitivity and specificity is defined as the least difference between the two,

as shown in Eq. (12). The threshold can also be specified as the one that produces the highest harmonic mean of the precision and recall (F1 score); however, this could result in a high disparity between sensitivity and specificity.

$$Threshold = \min_t |sensitivity(t) - specificity(t)| \quad (12)$$

5. Discussion

After training the model with the mentioned dataset and comparing the results with related papers, the proposed model, which leverages vectorization techniques such as embedding to represent drugs and proteins, proved to have better results in terms of MSE, CI, and r_m^2 . These results suggest that the model effectively captures the underlying relationships within the data due to the expressive power of vector representations compared to raw string values, which often fail to capture the intricacies of these molecules. Hence, instead of directly inputting the raw concatenated values, the drugs and proteins were rather represented in a latent space and saved before starting the training process; doing so enhances the model's performance and efficiency.

As for the results of [64,66] on the Kiba, DTC, and Metz datasets for the r_m^2 evaluation metric, their reported results were slightly higher compared to the mean of our results. However, when considering

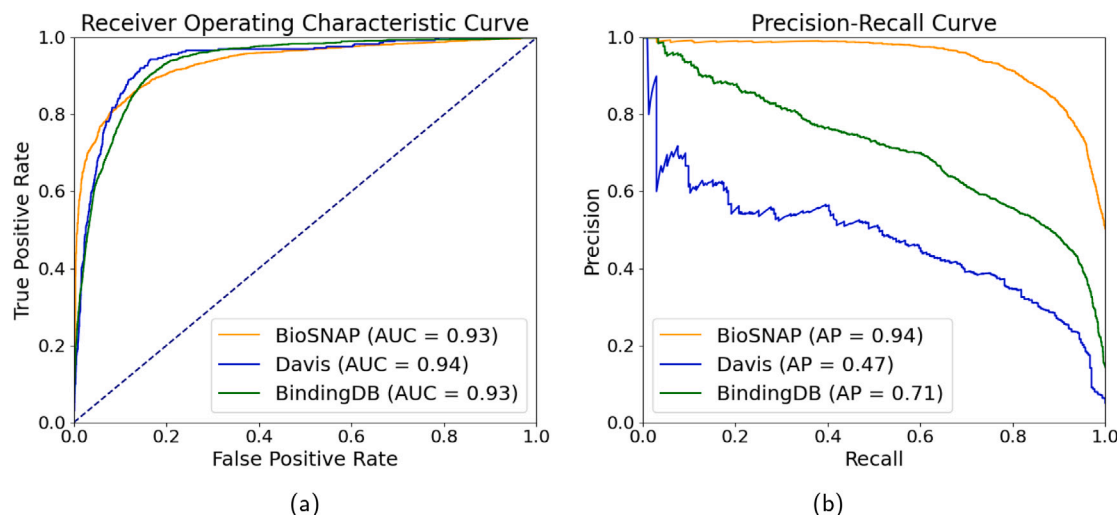


Fig. 6. Model performance on a random experiment for each classification dataset displaying (a) Receiver Operating Characteristic (ROC) curve and (b) Precision-Recall (PR) curve.

Table 12

Model performance on five experiments compared to related work for Davis classification dataset.

Paper	Sensitivity	Specificity	AUC	AUPRC	Threshold
LR	0.699 ± 0.051	0.842 ± 0.033	0.835 ± 0.010	0.232 ± 0.023	0.399
DNN	0.764 ± 0.045	0.860 ± 0.038	0.864 ± 0.009	0.258 ± 0.024	0.489
GNN-CPI	0.696 ± 0.047	0.842 ± 0.039	0.840 ± 0.012	0.269 ± 0.020	0.487
DeepDTI	0.751 ± 0.015	0.853 ± 0.012	0.861 ± 0.002	0.231 ± 0.006	0.387
DeepDTA	0.764 ± 0.045	0.865 ± 0.020	0.880 ± 0.007	0.302 ± 0.044	0.482
DeepConv-DTI	0.754 ± 0.040	0.880 ± 0.024	0.884 ± 0.008	0.299 ± 0.039	0.438
MolTrans	0.800 ± 0.022	0.876 ± 0.013	0.907 ± 0.002	0.404 ± 0.016	0.447
WAE-DTI	0.900 ± 0.013	0.867 ± 0.010	0.936 ± 0.002	0.474 ± 0.011	0.528

Table 13

Model performance on five experiments compared to related work for BindingDB classification dataset.

Paper	Sensitivity	Specificity	AUC	AUPRC	Threshold
LR	0.741 ± 0.013	0.896 ± 0.011	0.887 ± 0.002	0.557 ± 0.015	0.394
DNN	0.769 ± 0.028	0.914 ± 0.021	0.908 ± 0.003	0.613 ± 0.015	0.371
GNN-CPI	0.754 ± 0.015	0.903 ± 0.011	0.900 ± 0.004	0.578 ± 0.015	0.406
DeepDTI	0.651 ± 0.024	0.895 ± 0.023	0.844 ± 0.002	0.429 ± 0.005	0.06
DeepDTA	0.780 ± 0.035	0.915 ± 0.016	0.913 ± 0.003	0.622 ± 0.012	0.305
DeepConv-DTI	0.781 ± 0.015	0.905 ± 0.013	0.908 ± 0.004	0.611 ± 0.015	0.318
MolTrans	0.797 ± 0.005	0.896 ± 0.007	0.914 ± 0.001	0.622 ± 0.007	0.355
WAE-DTI	0.885 ± 0.006	0.853 ± 0.007	0.931 ± 0.001	0.707 ± 0.005	0.517

the standard deviation, the variability suggests that the higher results reported by them are within the achievable range of our model. According to the empirical rule, their results fall within one standard deviation of the mean, which includes approximately 68% of the data points, indicating that the difference is not statistically significant.

Also, it seems that the model is not intended to tackle novel drugs or proteins. If an unseen drug or protein is introduced in the testing set, the model is likely to provide poor predictions. This task is commonly referred to as zero-shot learning, where not only the interaction pair is unseen during testing, but also an unprecedented drug or protein, or both, is introduced during the testing phase. To overcome this challenge, the model must effectively capture the chemical properties of both the drug and the protein. Nevertheless, discovering novel drugs is becoming increasingly rare. However, the number of protein forms can be nearly infinite.

It is worth mentioning that some studies, such as [65], are not applicable for comparison with our proposed approach. Although they use similar datasets, there is a mismatch between the training and testing sets used. Specifically, the mentioned authors have a smaller number of examples. Therefore, directly comparing the approaches

is invalid and potentially unfair, as comparing results with different datasets does not provide conclusive indications. To address this issue, the number of examples in the dataset and the splitting strategy must be standardized to ensure a statistically sound comparison.

As future work, to enhance the robustness of our model, the properties of the Kiba, DTC, and Metz datasets must be analyzed and compared to the other datasets in terms of the r_m^2 evaluation metric. This analysis should include comparing metadata of both the drugs and proteins, such as atom and bond counts, drug weight, number of heavy atoms, number of hydrogen atoms, aromaticity of proteins, and molecular weight. To address the zero-shot learning problem, incorporating additional examples could help mitigate performance loss. Additionally, adding metadata for drugs and proteins as features can enable the model to better handle novel examples.

6. Conclusion

We have formulated a weighted average ensemble method to predict drug-target interactions, named WAE-DTI, which is based on vectorizing drugs using nine fingerprint descriptors, namely atom pair,

Table A.1

Models performance on five experiments prior to ensemble learning for Davis regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	0.229 ± 0.006	0.892 ± 0.002	0.706 ± 0.007	7.150 ± 1.506
Avalon fingerprint	ESM-2	0.213 ± 0.005	0.897 ± 0.004	0.718 ± 0.011	10.303 ± 0.661
MACCS keys fingerprint	ESM-2	0.217 ± 0.002	0.895 ± 0.004	0.706 ± 0.007	9.363 ± 1.365
MH fingerprint	ESM-2	0.217 ± 0.007	0.894 ± 0.002	0.709 ± 0.014	14.253 ± 1.849
Morgan fingerprint	ESM-2	0.220 ± 0.005	0.895 ± 0.003	0.700 ± 0.011	12.947 ± 1.476
RDKit fingerprint	ESM-2	0.222 ± 0.008	0.895 ± 0.002	0.711 ± 0.011	10.047 ± 2.675
SEC fingerprint	ESM-2	0.219 ± 0.004	0.893 ± 0.004	0.711 ± 0.004	11.993 ± 1.107
Topological torsion fingerprint	ESM-2	0.218 ± 0.007	0.896 ± 0.002	0.704 ± 0.012	12.587 ± 2.957
LDP	ESM-2	0.310 ± 0.003	0.853 ± 0.003	0.597 ± 0.014	18.023 ± 3.471

Table A.2

Models performance on five experiments prior to ensemble learning for Kiba regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	0.155 ± 0.002	0.885 ± 0.001	0.751 ± 0.006	55.157 ± 6.165
Avalon fingerprint	ESM-2	0.150 ± 0.002	0.883 ± 0.001	0.772 ± 0.006	49.700 ± 5.369
MACCS keys fingerprint	ESM-2	0.171 ± 0.002	0.868 ± 0.003	0.732 ± 0.013	58.030 ± 3.450
MH fingerprint	ESM-2	0.158 ± 0.002	0.883 ± 0.001	0.757 ± 0.014	57.120 ± 5.583
Morgan fingerprint	ESM-2	0.157 ± 0.003	0.883 ± 0.001	0.753 ± 0.014	56.350 ± 4.836
RDKit fingerprint	ESM-2	0.154 ± 0.001	0.885 ± 0.002	0.751 ± 0.010	63.560 ± 14.731
SEC fingerprint	ESM-2	0.158 ± 0.002	0.883 ± 0.002	0.756 ± 0.009	60.590 ± 16.958
Topological torsion fingerprint	ESM-2	0.158 ± 0.004	0.883 ± 0.003	0.747 ± 0.015	51.160 ± 12.123
LDP	ESM-2	0.389 ± 0.003	0.756 ± 0.001	0.422 ± 0.007	92.910 ± 0.624

Table A.3

Models performance on five experiments prior to ensemble learning for DTC regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	0.188 ± 0.004	0.883 ± 0.001	0.787 ± 0.009	28.917 ± 4.275
Avalon fingerprint	ESM-2	0.177 ± 0.003	0.883 ± 0.003	0.812 ± 0.013	22.953 ± 5.145
MACCS keys fingerprint	ESM-2	0.201 ± 0.003	0.867 ± 0.003	0.800 ± 0.004	22.570 ± 5.897
MH fingerprint	ESM-2	0.193 ± 0.006	0.877 ± 0.004	0.781 ± 0.010	33.233 ± 4.758
Morgan fingerprint	ESM-2	0.191 ± 0.004	0.880 ± 0.002	0.789 ± 0.015	24.147 ± 3.512
RDKit fingerprint	ESM-2	0.179 ± 0.005	0.885 ± 0.002	0.806 ± 0.017	29.973 ± 4.307
SEC fingerprint	ESM-2	0.191 ± 0.001	0.881 ± 0.001	0.789 ± 0.007	24.033 ± 3.985
Topological torsion fingerprint	ESM-2	0.189 ± 0.003	0.880 ± 0.001	0.786 ± 0.007	28.167 ± 2.972
LDP	ESM-2	0.434 ± 0.005	0.768 ± 0.001	0.565 ± 0.007	46.150 ± 2.969

Avalon, MACCS keys, MH, Morgan, RDKit, SEC, topological torsion, and LDP, in addition to using ESM-2 as a protein embedder. We have persisted the vectors for all descriptors beforehand to prevent recalculation during each epoch. Also, we have assigned each of the nine models a weight based on its performance on the evaluation set, the model's weights are then used for prediction using a weighted average ensemble method. We have implemented a punishment factor to suppress underperforming models. The models are trained on six datasets independently: Davis, KIBA, DTC, Metz, ToxCast, and STITCH. To further evaluate the models generalizability, we have adapted them to perform classification tasks on the BioSNAP, Davis, and BindingDB datasets. The results prove that our proposed model provided tangible improvements across all datasets. To further enhance the results, we propose conducting an error analysis to identify and study the areas where the model had high prediction errors.

CRediT authorship contribution statement

Tariq Sha'ban: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Ahmad M. Mustafa:** Writing – review & editing, Supervision, Conceptualization. **Mostafa Z. Ali:** Supervision, Conceptualization.

Ethical statement

This research did not involve the collection of any new data from human or animal subjects. All analyses and findings presented in this study are based on publicly available datasets. As a result, there was no need for ethical approval from a review board. We affirm that no ethical concerns are applicable to this work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Regression results prior to using ensemble learning

Tables A.1, A.2, A.3, A.4, A.5, and A.6 show the mean performance and standard deviations of five repeated experiments for each trained model prior to their use in the ensemble model. In addition to incorporating the MSE, CI, and r_m^2 of the test set, the table also displays the time in minutes required for the model to complete the training process, which might increase significantly on larger datasets and methods with large vector output. Also, models which provide poor representations may take longer to converge suboptimally.

Appendix B. Classification results prior to using ensemble learning

Similarly to the tables in Appendix A, Tables B.1, B.2, and B.3 show the mean performance and standard deviations of five repeated experiments for each trained model prior to their use in the ensemble model on the classification task. However, the MH fingerprint descriptor was omitted from the results due to its inferior performance.

Table A.4

Models performance on five experiments prior to ensemble learning for Metz regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	0.357 ± 0.004	0.787 ± 0.002	0.570 ± 0.005	7.623 ± 1.744
Avalon fingerprint	ESM-2	0.325 ± 0.009	0.798 ± 0.005	0.616 ± 0.005	11.030 ± 5.194
MACCS keys fingerprint	ESM-2	0.337 ± 0.008	0.793 ± 0.003	0.610 ± 0.005	8.653 ± 1.907
MH fingerprint	ESM-2	0.358 ± 0.008	0.787 ± 0.002	0.575 ± 0.016	11.217 ± 3.598
Morgan fingerprint	ESM-2	0.365 ± 0.003	0.785 ± 0.001	0.561 ± 0.004	6.803 ± 0.760
RDKit fingerprint	ESM-2	0.330 ± 0.003	0.796 ± 0.001	0.610 ± 0.008	9.240 ± 1.833
SEC fingerprint	ESM-2	0.364 ± 0.005	0.785 ± 0.002	0.563 ± 0.009	6.617 ± 2.395
Topological torsion fingerprint	ESM-2	0.363 ± 0.002	0.787 ± 0.001	0.563 ± 0.006	10.443 ± 3.239
LDP	ESM-2	0.532 ± 0.009	0.723 ± 0.003	0.418 ± 0.011	14.147 ± 3.052

Table A.5

Models performance on five experiments prior to ensemble learning for ToxCast regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	0.326 ± 0.002	0.914 ± 0.002	0.553 ± 0.008	37.240 ± 1.505
Avalon fingerprint	ESM-2	0.321 ± 0.003	0.916 ± 0.001	0.559 ± 0.009	29.150 ± 0.219
MACCS keys fingerprint	ESM-2	0.324 ± 0.000	0.914 ± 0.001	0.561 ± 0.006	31.267 ± 0.694
MH fingerprint	ESM-2	<i>high value</i>	0.467 ± 0.030	0.004 ± 0.005	31.363 ± 1.173
Morgan fingerprint	ESM-2	0.333 ± 0.002	0.911 ± 0.002	0.540 ± 0.012	34.947 ± 1.226
RDKit fingerprint	ESM-2	0.323 ± 0.002	0.915 ± 0.001	0.553 ± 0.005	36.857 ± 1.719
SEC fingerprint	ESM-2	0.333 ± 0.003	0.911 ± 0.002	0.540 ± 0.009	35.323 ± 0.961
Topological torsion fingerprint	ESM-2	0.331 ± 0.002	0.913 ± 0.001	0.541 ± 0.006	36.687 ± 2.912
LDP	ESM-2	0.369 ± 0.001	0.898 ± 0.000	0.506 ± 0.002	69.043 ± 10.728

Table A.6

Models performance on five experiments prior to ensemble learning for STITCH regression dataset.

Drug	Protein	MSE	CI	r_m^2	mins
Atom pair fingerprint	ESM-2	1.140 ± 0.003	0.751 ± 0.004	0.389 ± 0.004	199.053 ± 8.135
Avalon fingerprint	ESM-2	1.078 ± 0.006	0.740 ± 0.006	0.417 ± 0.003	303.273 ± 86.829
MACCS keys fingerprint	ESM-2	1.120 ± 0.006	0.708 ± 0.004	0.407 ± 0.002	250.583 ± 58.438
MH fingerprint	ESM-2	<i>high value</i>	0.504 ± 0.022	0.001 ± 0.001	164.047 ± 1.148
Morgan fingerprint	ESM-2	1.122 ± 0.006	0.763 ± 0.004	0.398 ± 0.005	183.923 ± 4.397
RDKit fingerprint	ESM-2	1.143 ± 0.008	0.743 ± 0.005	0.389 ± 0.006	262.890 ± 35.466
SEC fingerprint	ESM-2	1.167 ± 0.006	0.759 ± 0.005	0.374 ± 0.007	186.147 ± 6.879
Topological torsion fingerprint	ESM-2	1.163 ± 0.003	0.754 ± 0.004	0.378 ± 0.004	196.000 ± 8.108
LDP	ESM-2	1.542 ± 0.003	0.627 ± 0.002	0.201 ± 0.001	300.370 ± 62.823

Table B.1

Models performance on five experiments prior to ensemble learning for BioSNAP classification dataset.

Drug	Protein	Sensitivity	Specificity	AUC
Atom pair fingerprint	ESM-2	0.824 ± 0.008	0.830 ± 0.019	0.899 ± 0.005
Avalon fingerprint	ESM-2	0.859 ± 0.008	0.872 ± 0.005	0.932 ± 0.002
MACCS keys fingerprint	ESM-2	0.873 ± 0.006	0.884 ± 0.006	0.942 ± 0.002
Morgan fingerprint	ESM-2	0.789 ± 0.008	0.830 ± 0.004	0.880 ± 0.003
RDKit fingerprint	ESM-2	0.841 ± 0.002	0.857 ± 0.009	0.915 ± 0.003
SEC fingerprint	ESM-2	0.795 ± 0.006	0.826 ± 0.011	0.879 ± 0.001
Topological torsion fingerprint	ESM-2	0.807 ± 0.014	0.834 ± 0.010	0.888 ± 0.004
LDP	ESM-2	0.767 ± 0.010	0.805 ± 0.012	0.858 ± 0.003
Drug	Protein	AUPRC	Threshold	mins
Atom pair fingerprint	ESM-2	0.905 ± 0.002	0.560 ± 0.251	7.083 ± 1.940
Avalon fingerprint	ESM-2	0.933 ± 0.001	0.582 ± 0.121	6.213 ± 1.021
MACCS keys fingerprint	ESM-2	0.945 ± 0.002	0.557 ± 0.126	5.757 ± 1.746
Morgan fingerprint	ESM-2	0.892 ± 0.004	0.701 ± 0.082	3.987 ± 0.970
RDKit fingerprint	ESM-2	0.917 ± 0.003	0.670 ± 0.186	7.670 ± 1.473
SEC fingerprint	ESM-2	0.888 ± 0.002	0.682 ± 0.053	4.057 ± 1.638
Topological torsion fingerprint	ESM-2	0.896 ± 0.003	0.442 ± 0.098	3.780 ± 0.970
LDP	ESM-2	0.862 ± 0.001	0.537 ± 0.047	1.820 ± 0.190

Table B.2

Models performance on five experiments prior to ensemble learning for Davis classification dataset.

Drug	Protein	Sensitivity	Specificity	AUC
Atom pair fingerprint	ESM-2	0.892 ± 0.009	0.853 ± 0.021	0.929 ± 0.005
Avalon fingerprint	ESM-2	0.882 ± 0.013	0.847 ± 0.005	0.925 ± 0.005
MACCS keys fingerprint	ESM-2	0.890 ± 0.006	0.862 ± 0.009	0.929 ± 0.003
Morgan fingerprint	ESM-2	0.881 ± 0.019	0.855 ± 0.017	0.926 ± 0.003
RDKit fingerprint	ESM-2	0.876 ± 0.020	0.844 ± 0.011	0.924 ± 0.004
SEC fingerprint	ESM-2	0.879 ± 0.018	0.852 ± 0.009	0.926 ± 0.004
Topological torsion fingerprint	ESM-2	0.874 ± 0.022	0.851 ± 0.007	0.919 ± 0.005
LDP	ESM-2	0.797 ± 0.015	0.728 ± 0.014	0.827 ± 0.003
Drug	Protein	AUPRC	Threshold	mins
Atom pair fingerprint	ESM-2	0.408 ± 0.014	0.559 ± 0.168	3.017 ± 0.508
Avalon fingerprint	ESM-2	0.431 ± 0.014	0.457 ± 0.082	1.660 ± 0.343
MACCS keys fingerprint	ESM-2	0.415 ± 0.021	0.564 ± 0.119	1.083 ± 0.072
Morgan fingerprint	ESM-2	0.390 ± 0.016	0.590 ± 0.289	2.580 ± 0.507
RDKit fingerprint	ESM-2	0.416 ± 0.002	0.485 ± 0.093	2.793 ± 0.230
SEC fingerprint	ESM-2	0.398 ± 0.022	0.561 ± 0.125	2.553 ± 0.299
Topological torsion fingerprint	ESM-2	0.383 ± 0.009	0.618 ± 0.048	2.480 ± 0.287
LDP	ESM-2	0.239 ± 0.003	0.537 ± 0.059	0.837 ± 0.113

Table B.3

Models performance on five experiments prior to ensemble learning for BindingDB classification dataset.

Drug	Protein	Sensitivity	Specificity	AUC
Atom pair fingerprint	ESM-2	0.869 ± 0.009	0.830 ± 0.015	0.909 ± 0.011
Avalon fingerprint	ESM-2	0.883 ± 0.007	0.834 ± 0.012	0.918 ± 0.003
MACCS keys fingerprint	ESM-2	0.886 ± 0.009	0.841 ± 0.006	0.923 ± 0.001
Morgan fingerprint	ESM-2	0.831 ± 0.005	0.794 ± 0.003	0.879 ± 0.002
RDKit fingerprint	ESM-2	0.885 ± 0.004	0.832 ± 0.007	0.914 ± 0.001
SEC fingerprint	ESM-2	0.844 ± 0.012	0.795 ± 0.006	0.881 ± 0.004
Topological torsion fingerprint	ESM-2	0.876 ± 0.015	0.822 ± 0.014	0.908 ± 0.013
LDP	ESM-2	0.818 ± 0.013	0.788 ± 0.019	0.873 ± 0.006
Drug	Protein	AUPRC	Threshold	mins
Atom pair fingerprint	ESM-2	0.636 ± 0.014	0.481 ± 0.054	6.157 ± 1.244
Avalon fingerprint	ESM-2	0.651 ± 0.005	0.457 ± 0.045	2.620 ± 0.413
MACCS keys fingerprint	ESM-2	0.671 ± 0.004	0.372 ± 0.082	3.777 ± 1.089
Morgan fingerprint	ESM-2	0.597 ± 0.001	0.468 ± 0.052	3.543 ± 0.475
RDKit fingerprint	ESM-2	0.634 ± 0.007	0.490 ± 0.076	3.733 ± 0.308
SEC fingerprint	ESM-2	0.598 ± 0.002	0.500 ± 0.049	3.360 ± 0.377
Topological torsion fingerprint	ESM-2	0.615 ± 0.016	0.518 ± 0.056	4.443 ± 1.776
LDP	ESM-2	0.511 ± 0.008	0.489 ± 0.057	2.673 ± 0.704

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